

Mohamed Shereef CH

+91-9061139031 mhdshareefch@gmail.com github.com/spigelspike linkedin.com/in/mohamed-shereef

EDUCATION

- Bachelor of Technology in Information Technology** 05/2026
MEA Engineering College, Kerala

TECHNICAL SKILLS

- Languages:** Python, JavaScript, TypeScript, Java, C
- Backend:** FastAPI, Node.js, REST APIs, JWT Authentication, Supabase Edge Functions, Async I/O
- Frontend:** React, Tailwind CSS, Vite, Zustand, HTML5/CSS3
- Databases:** PostgreSQL (RLS, indexing), Supabase (Realtime, Auth), Firebase Firestore
- AI/ML:** PyTorch, EfficientNet, Grad-CAM, Hugging Face, Ollama
- Tools:** Git, Docker, Linux/Bash, Postman, Vercel, ElevenLabs, Ably, Agora

PROJECTS

- Kallanum Policeum — Realtime Multiplayer Social Deduction Game** React, TypeScript, Supabase, Ably, Agora |
Live: kallanumpoliceum.online | GitHub
 - Shipped a production serverless multiplayer game supporting up to 15 concurrent players per room, with all game logic in Supabase Edge Functions — eliminating dedicated server infrastructure entirely
 - Achieved real-time game state sync across all clients via Supabase Realtime; integrated Ably for persistent world chat and Agora for low-latency voice, both authenticated using server-generated tokens to prevent client spoofing
 - Engineered server-side role assignment to prevent client-side cheating, MCQ deduction mechanics with bluff zones, room codes via `crypto.randomBytes()` to resist brute-force guessing, and a Web Audio API sound system
 - Secured all database tables with PostgreSQL RLS, enforced JWT on every Edge Function, and hardened the client with rate limiting, CSP headers, and CORS — no player can read or spoof another's game state
- Book2Vision — AI Multimedia Generation Platform** Python, FastAPI, Gemini, ElevenLabs, Pollinations.ai | GitHub
 - Engineered an end-to-end pipeline converting full-length books into chapter-level audiobooks and visual summaries via an OCR → NLP → LLM → TTS processing chain
 - Decoupled 30–120 second generation tasks (OCR, LLM inference, TTS rendering) from the HTTP request lifecycle using FastAPI background tasks, eliminating client timeouts on long-running jobs
 - Abstracted ElevenLabs, Gemini, and Pollinations.ai behind a shared service layer with automatic provider fallback (e.g., Edge TTS when ElevenLabs quota is hit) without changing caller code
- ACLInsight-Net — Deep Learning MRI Diagnostic System** Python, PyTorch, EfficientNet-B4, Grad-CAM, MRNet | GitHub
 - Trained an EfficientNet-B4 multi-plane fusion model (axial, coronal, sagittal) for ACL tear detection on the MRNet dataset (1,250 knee MRI studies)
 - Achieved AUC-ROC of 0.9512 and 90.83% accuracy with 5-fold cross-validation; applied Grad-CAM to generate spatial heatmaps identifying ACL-region activations for clinical interpretability
 - Resolved CUDA OOM crashes via gradient checkpointing and mixed-precision (float16) training; fixed RAM overflow with lazy dataset loading; restored interrupted runs via epoch-level checkpointing; submitted to IEEE
- CPMS — Campus Placement Management System** JavaScript, PostgreSQL, Supabase, Gemini | GitHub
 - Developed a role-based campus placement platform handling the complete pipeline from student profile submission and document upload through recruiter shortlisting and offer dispatch, across 3 distinct user roles
 - Designed a normalized PostgreSQL schema with Row-Level Security policies ensuring students, recruiters, and admins each access strictly their permitted records
 - Implemented JWT session management, paginated REST endpoints, and PDF resume parsing with structured field extraction
 - Integrated Gemini 1.5 Flash (OpenRouter fallback) to auto-score uploaded resumes against job descriptions, reducing manual shortlisting effort per batch

OPEN SOURCE CONTRIBUTIONS

- OmniVoice TTS** — Built `ml_normalize.py` (Malayalam text normalization: numbers, currency, dates, units) for the open-source TTS library; contributed a FastAPI demo and Colab notebook alongside the merged PR Python | PR Merged

CERTIFICATIONS & ACHIEVEMENTS

- Complete 2026 Python Bootcamp: Learn Python from Scratch** 07/2026
CodeWithHarry
- The Complete Full-Stack Web Development Bootcamp** 06/2025
Udemy — Dr. Angela Yu
- Kaggle Hackathon — Avengers: Doomsday (Network Intrusion Classification)** 2025
Ranked #5 · Score: 0.9824

LEADERSHIP & ACTIVITIES

- Technical Lead**, Xorir — IT Association, MEA Engineering College (2024–2026). Organized technical workshops and peer mentoring sessions for 40+ students; coordinated annual department technical events.
- Active Member**, MEA FOSS — Open-Source Club (2023–2026). Delivered sessions on FastAPI backend architecture; guided members through open-source contribution workflows.

ADDITIONAL INFORMATION

- Languages:** English (professional), Malayalam (native), Hindi